MILITARY SPECIFICATION

DEHUMIDIFIERS, ROTARY, DESICCANT, ELECTRIC

(FOR SHIPBOARD USE)

1. SCOPE

1.1 Scope. This specification covers rotary electric solid desiccant dehumidifiers.

1.2 Classification. Dehumidifiers shall be of the following sizes as specified (see 6.2).

20-153
36-255
55-505

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

TT-P-645	- Primer, Paint, Zinc-Chromate, Alkyd Type.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

MIL-S-901	- Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-E-917	- Electric Power Equipment, Basic Requirements (Naval Shipboard Use).
MIL-D-1000	- Drawings, Engineering and Associated Lists.
MIL-D-1000/2	- Drawings, Engineering and Associated Lists.
MIL-C-2212	- Controllers, Alternating-Current, Naval Shipboard.
MIL-M-9868	- Microfilming of Engineering Documents, 35 MM, Requirements for.
MIL-M-9868/1	- Microfilming of Engineering Documents, 35 MM, for Naval Ship Systems.
MIL-P-15024	- Plates Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071	- Manuals, Equipment and Systems.
MIL-E-15090	- Enamel, Equipment, Light-Gray (Formula No. 111).
MIL-P-15137	- Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15280	- Plastic Foam, Unicellular, Sheet and Tubular Form, Elastomeric.
MIL-P-15328	- Primer, Pretreatment (Formula No. 117 for Metals).
MIL-E-15599	- Electrodes, Welding, Covered: Low and Medium-Carbon Steel.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-F-16552	- Filters, Air Environmental Control System, Cleanable Impingement (High Velocity Type).
MIL-M-17059	- Motors, 60-Cycle, Alternating-Current, Fractional H. P., (Shipboard Use).
MIL-M-17060	- Motors, 60 Cycle, Alternating Current, Integral Horsepower (Shipboard Use).
MIL-P-17545	- Primer Coating, Steel Ship Maintenance, Alkyd-Red Lead Type, Formula No. 116.
MIL-I-22023	- Insulation Felt, Thermal and Sound Absorbing Felt, Fibrous Glass, Flexible.
MIL-H-22577	- Heating Elements, Electrical; Cartridge, Strip and Tubular Type.

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STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Fabrication, Welding and Inspection of Machinery, Piping and Pressure Vessels for Ships of the United States Navy.
MIL-STD-804 - Formats and Coding of Aperture, Copy and Tabulating Cards for Engineering Data Micro-Reproduction System.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on data of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY OF HEATING, REFRIGERATION, AND AIR-CONDITIONING ENGINEERS (ASHRAE)

36-62 - Measurement of Sound Power Radiated from Heating, Refrigerating, and Air Conditioning Equipment, Criteria for Testing.

(Application for copies should be addressed to the American Society of Heating, Refrigerating and Air-Conditioning Engineers, United Engineering Center, 345 East 47th Street, New York, New York 10017).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A-164 - Electrodeposited Coatings of Zinc on Steel.
A-165 - Electrodeposited Coatings of Cadmium on Steel.
A-386 - Zinc Coating (Hot Dip) on Assembled Steel Products.
B-6 - Slab Zinc.
B-201 - Chromate Finishes on Electrodeposited Zinc, Hot-Dipped Galvanized, and Zinc Die-Cast Surfaces.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups using Federal agencies.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Ill. 60606.)

3. REQUIREMENTS

3.1 Preproduction. Prior to beginning production, a sample shall be inspected as specified in 4.3.

3.2 Materials. All materials forming a part of the finished product shall be suitable for the purpose intended. The materials shall be free from any defects that might affect the serviceability or appearance of the finished product.

3.2.1 Corrosion protection. Corrosion-resisting steel, galvanized steel, copper and brass referenced herein are considered satisfactory corrosion-resisting materials. Corrosion-resisting steel when fabricated by any method that tends to reduce corrosion-resisting properties shall be normalized to restore those properties before being assembled in any unit.

3.2.1.1 Parts fabricated from other than corrosion-resisting materials shall be protected against corrosion after fabrication with chemicals, electrolytic processes, plating, or suitable paints. The following methods while not restrictive are considered corrosion protection methods when properly applied.

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(a) Hot-dipped galvanized in accordance with ASTM A-386 with the spelter conforming to grade 5 of ASTM B-6.
(b) Electroplating with zinc in accordance with type LS of ASTM A-164 followed by a chromate treatment conforming to ASTM B-201.
(c) Electroplating with cadmium in accordance with type NS of ASTM A-165 followed by a chromate treatment conforming to ASTM B-201.
(d) Hot phosphoric or chromic acid treatment, or a coating of primer in accordance with MIL-P-15328 followed by two coats of primer conforming to TT-P-664.

3.2.1.2 All bolts, nuts, studs, screws and such fastening or fittings as may be used shall be of corrosion-resisting material, or of a material treated in a manner to render it adequately resistant to corrosion. Self-tapping screws with sheet metal threads shall not be used.

3.2.2 Dissimilar metals. To prevent destructive electrolysis, direct contact of electrolytically dissimilar materials shall be avoided as far as practicable.

3.3 Design and construction.

3.3.1 General. Each dehumidifier shall consist of the desiccant contactor (wheel), a conditioned-air (dry air) fan and motor, a reactivation fan and motor, a reactivation heater, air filters and drive system, mounted and enclosed in a metal cabinet provided with air passages, air seals, safety devices, controls and thermal insulation, designed for deck mounting and incorporating a provision to enable connection to supply ducts and to discharge ducts and ready for installation and connection to a power source.

3.3.1.1 General shipboard design conditions. General shipboard design characteristics shall be as follows:

(a) Capable of operating for at least 50,000 hours with up to 50,000 operating cycles and without major overhaul of less than 6,000 hours intervals.
(b) Perform in accordance with requirements herein under ambient design temperature of 0°F. minimum to 120°F. maximum.
(c) Capable of operating satisfactorily on the weather deck of a ship when exposed to soaking by rain and extreme atmospheric conditions.
(d) Operate satisfactorily and maintain adequate lubrication when permanently inclined 15 degrees from the normal horizontal position in any direction.

3.3.1.2 Shock. When specified (see 6.2), the dehumidifier shall be designed such that it is capable of passing the high impact shock tests specified in MIL-S-901 for grade A, class I equipment.

3.3.1.3 Noise. When specified (see 6.2), the assembly by design and provision of noise treatment shall meet the sound power level indicated herein. The noise level shall be measured as specified in 4.3.3.3. The measured noise level shall be such that the summation of deviations from the specified limit shall be equal to zero or a negative value as shown in table I.

Table I - Sound power level, decibels re 10^{-13} watts

Octave band (CPS)	37.5	75 150	150 300	300 600	600 1200	1200 2400	2400 4800	4800 10KC
Size 20-153	101	98	95	92	89	86	84	82
Size 36-255	102	99	96	93	90	87	85	83
Size 55-505	104	101	98	95	92	89	87	85

3.3.1.4 Reliability and maintainability. To meet the needs of the Naval service, it is imperative that reliability of operation and maintainability be considered of prime importance in the design and manufacture of the dehumidifier. Human engineering shall be considered in design to preclude or minimize possibility of failure through improper or poor maintenance.

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3.3.1.4.1 The supplier shall provide and maintain an effective reliability and maintainability design assurance program including the following:

(a) Design review. The program shall include provisions for the review and evaluation of designs as an integral part of the supplier's engineering design procedures. Design or engineering changes occurring during development or production shall be subjected to comparable review procedures.

(b) Reliability analysis. To indicate the methods used to achieve reliability, the supplier shall submit an analysis of those factors affecting reliability. The analysis shall include, but shall not be limited to the following:

(1) List of those parts which experience and judgment show are subject to wear, material deterioration, and service failures.

(2) Specific design features employed to attain the required service life of the parts with due consideration of shipboard environment and resultant conditions. Some suggested design features are choice of materials, compatibility of materials, hardness, surface finishes, fits, clearances, fastenings, repairability and accessibility.

(3) Preventive maintenance and servicing requirements necessary to the achievement of reliable equipment. Any unusual steps or precautions necessary in carrying out maintenance and servicing requirements shall be emphasized.

(c) Failure reporting, analysis and feedback. The program shall incorporate a formalized system for the recording, collecting, and analysis of all failures that occur during all testing, installation and operation through the tenure of the contract. Analysis of all failures shall be fed back to the supplier's engineering, management and production activities on a timely basis. Failure reports received from using activity shall be integrated into this program for trouble analysis and for experience considerations for future design review.

3.3.1.4.2 Maintainability of the equipment shall be demonstrated as specified in 4.2.3.

3.3.2 Capacity rating. The capacity rating of the dehumidifier shall be at least equal to that specified in table II, based on 70°F. dry-bulb temperature (DB) and 35 percent relative humidity (RH) entering the conditioned and reactivation inlets. The airflow at the conditioned air outlet duct connection shall be within plus and minus 10 percent of the rated air flow specified in table II. The economy of operation at rated air flow and air temperature and humidity conditions shall not exceed 1.10 kilowatt-hour per pound of water removal.

Table II - Capacity

Dehumidifier Sizes	Airflow S.C.F.M.	Rate of water removal Pounds per hour
20-153	150	2.0
36-255	250	3.6
55-505	500	5.5

3.3.2.1 Performance. The dehumidifier shall be noncycling, that is, it shall produce dry air of constant psychrometric quality for a given inlet air condition. The performance of the dehumidifier at rated air flow shall be as specified in table III. The economy of operation shall not exceed that specified in table III, and the water removal rate shall be at least equal to that specified in table III.

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Table III - Performance

Inlet air conditions		Minimum rate of water removal in pounds per hour			Power (maximum) in kilowatt - hour per pound of water removal
Temperature °F	Humidity percent	Size			
		20-153	36-255	55-505	
95	35	2.7	4.8	7.5	0.80
40	35	1.0	1.7	2.6	1.60

3.3.3 Physical dimensions and weight. The design of the unit shall be such that space and weight are kept to a minimum.

3.3.3.1 Physical dimensions. The overall dimensions of any dehumidifier shall be not greater than 25 inches in width, 60 inches in depth and 53 inches in height. Dehumidifiers greater than 48 inches in depth shall have disassembly features such that it can be turned in a 48 inch passageway.

3.3.3.2 Operating weight. The operating weight shall not exceed 150 pounds for the size 20-153 dehumidifier, 450 pounds for the size 36-255 dehumidifier and 600 pounds for the size 55-505 dehumidifier.

3.3.4 Cabinet. The cabinet enclosure shall be constructed of steel or aluminum. The frame or chassis shall be rigid and of adequate strength to support and maintain alignment of the assembled parts. Provision shall be made to permit access to and removal of desiccant contactor, air seals, fans, heater and electric motors through removable side panels. Air filters shall be provided separate access openings. The removable side panels shall be interchangeable on each assembly. Surfaces which mate removable panels and access openings shall be gasketed to produce a weathertight seal.

3.3.4.1 Air flow connections. Air flow connections shall be provided for a humid air inlet, a conditioned (dry) air outlet, the reactivation air inlet and the reactivation air outlet. The inlets and outlets shall have smooth flat surfaces for flanged duct connections. The dehumidifier shall be delivered with the openings closed with blank flanges and gaskets.

3.3.4.1.1 Connection sizes. The dimensions of the reactivation air inlet connection shall be 3 inches in diameter for the size 20-153, 4 inches in diameter for the size 36-255 and 9 by 9 inches for the size 55-505 dehumidifier. The dimensions of the reactivation air outlet connection shall be 3 inches in diameter for size 20-153, 4 inches in diameter for the size 36-255 and 3-1/2 by 7-1/8 inches for the size 55-505 dehumidifier. The dimensions of the humid air inlet connections shall be 7 by 7-1/8 inches for the size 20-153, 8-1/2 by 8-1/2 inches for the size 36-255 and 8 by 18 inches for the size 55-505 dehumidifier. The dimensions of the dry air outlet connection shall by 6 by 6 inches for the size 20-153, 7 by 7 inches for the size 36-255 and 3-1/2 by 7-1/8 inches for the size 55-505 dehumidifier. The dimensions of each connection shall not vary more than plus or minus 1/8 inch from that specified.

3.3.4.2 Lifting pads. The cabinet shall have at least four pad eyes, eye bolts, lifting pads, or handles for moving the dehumidifier through a passageway or onto a foundation.

3.3.5 Desiccant contactor. The desiccant contactor shall consist of nongranular fixed desiccant impregnated structure incapable of producing flammable or noxious gases at any temperature within the design operating temperature range and up to 510°C (950°F). If the desiccant contactor is subjected to a temperature equal to set point of the high temperature cutout (see 3.7.2), the desiccant contactor shall recover full performance when returned to design operating temperatures. The desiccant contactor shall be supported such that self-grinding, settling, erosion or attrition will not result from air friction, thermal stresses, or other conditions which can be encountered for normal equipment operation. At design velocity the static pressure loss shall be not greater than 2.0 inches of water through the desiccant contactor. The desiccant shall be of adequate capacity to achieve the performance requirements of 3.3.2.1, but it shall not produce free liquid. The desiccant shall be capable of repeated reactivation. There shall be no measurable weight

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of desiccant carryover to the dry air (conditioned air) outlet at maximum air flow provided the inlet air dew-point does not exceed 85.5°F (188 grains per pound of dry air).

3.3.5.1 Drive system. The desiccant contactor (wheel) shall be provided with a V-belt drive system which does not require routine lubrication to insure consistent performance with no variations. A positive acting belt adjustment device shall be provided to permit adjustment of belt tension.

3.3.5.2 Air streams. The conditioned air (humid-dry air) stream and the reactivation air stream shall be counter-flow through the desiccant contactor (wheel) to accomplish self-cleaning and thermal efficiency.

3.3.5.3 Connections for air passages. Connections in all air passages shall be of welded construction, by bolted flanges or any other method that is acceptable to the command or agency concerned. Flanged connections shall be gasketed and air tight. Gaskets shall be of asbestos or of material capable of withstanding intermittent temperature of 500°F.

3.3.6 Air seals. Air seals shall be provided to prevent air from bypassing the desiccant contactor and to prevent air leakage between the reactivation and the condition air circuits. Seal material shall be selected to obtain maximum performance reliability specified in 3.3.1.4.1 and conforming to 3.3.1.1(a). The air seals shall be so constructed that they will be long-wearing and not cause damage to the desiccant contactor. The air seals shall be removable and shall be adjustable where required to prevent air leakage and air from bypassing the desiccant contactor (see 4.2.3).

3.4 Air filters. Air filters shall be provided for filtering all air entering the desiccant contactor (wheel). The performance of the air filters shall conform to MIL-F-16552. All air filters shall be of a size such that the face velocity of the air filter is at least 450 feet per minute.

3.5 Thermal insulation. Surfaces of the dehumidifier that are subject to the formation of condensate shall be provided with insulation to prevent dripping or running off of moisture under rated capacity conditions. Surfaces of the dehumidifier which can attain a temperature of 130°F. or higher during any service condition shall be insulated wherever necessary to protect personnel, prevent undesirable transfer of heat to the surroundings, or prevent transfer of heat from the component wherever such transfer would be detrimental to operation of the component or system. The insulation shall be in accordance with type II of MIL-P-15280 and type I, class 6 of MIL-I-22023, as applicable.

3.6 Fans. The conditioned (dry) air fan and the reactivation (re-generator) fan of the dehumidifier shall be of the centrifugal type, complete with motor. The fans shall be direct-driven units with the fan wheels overhung on the motor shafts. The fans shall be quiet in operation and shall be statically and dynamically balanced. The fan housings shall have smooth curved inlets, and all rotating parts of the fans shall be designed to have a factor of safety of not less than 8, based on the ultimate tensile strength of the material involved. The fan sheels and housings shall be constructed of steel or aluminum.

3.6.1 Fan wheels. The fan wheel of each fan shall be of the multi-blade type and shall be smooth and free from projecting bolt heads and nuts or other projections which might catch dirt and lint or cause noisy fan operation.

3.6.2 Conditioned (dry) air fan. Unless otherwise specified (see 6.2), the fan shall produce an available total discharge pressure of 5 inches of water at the conditioned (dry) air outlet of the size 36-255 and size 55-505 dehumidifier and 3 inches of water at the conditioned (dry) air outlet of the size 20-153 dehumidifier.

3.6.3 Reactivation (regenerator) fan. Unless otherwise specified (see 6.2), the fan of any dehumidifier shall produce an available total discharge pressure of 1 inch of water at the reactivation air outlet. The fan shall be located downstream of the regenerator section of the desiccant contactor (wheel).

3.7 Electrical requirements.

3.7.1 Dehumidifier total connected load. The dehumidifiers total connected electrical loads shall not exceed 4.6 kilowatts (kw.) for the size 20-153, 7.5 kw. for the size 36-255 and 11.5 kw. for size 55-505 at rated voltage.

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3.7.2 Control arrangements. Circuits shall be arranged and protective devices shall be provided to insure full protection of all electrical devices. All such protective devices shall automatically reset when the need for protection ceases. Such devices shall remain in proper adjustment under all normal conditions of operation and after repeated functioning as a safety device. In particular the following shall be provided in addition to any other protection which may be necessary:

(a) Low voltage release for all motors.
(b) Means to insure that reactivation heater cannot be energized before the reactivation fan has started, and that the reactivation fan continues to operate for at least two minutes after the heater is de-energized.
(c) Means to insure that reactivation heater is de-energized when temperature of regenerator air exceeds design limitations.
(d) The reactivation fan and heater are to operate for at least two minutes before the conditioned air (dry air) fan is started after a period of idleness.
(e) Overload relays shall be provided for the protection of each fan motor.
(f) Thermal overloads shall be sized to protect the associated equipment in an ambient temperature of 50°C (122°F).
(g) Protective thermostat in the reactivation air outlet to de-energize dehumidifier in the event the air temperature exceeds 500°F.

3.7.3 Electrical equipment. The design of the electrical equipment shall be in general accordance with MIL-E-917.

3.7.3.1 Electric service. The fan motors, controller and regenerator heater shall be supplied for 440 volts, three phase, 60 cycle alternating current (ac) service.

3.7.3.2 Motors. Unless otherwise specified (see 6.2), all motors shall be service C in accordance with MIL-M-17059 or MIL-M-17060, as applicable. The motors shall be designed for 50°C. (122°F.) ambient temperature, continuous duty, constant speed, and spraytight enclosure and shall be sufficiently sized (standard horsepower size) to meet the fan or drive system load requirements, as applicable. All the motors shall be equipped with ball bearings.

3.7.3.3 Controller. When specified (see 6.2), an external (remote) controller shall be provided for energizing the system. The controller shall be in accordance with MIL-C-2212 and the following classification:

Ambient	- 50°C.
Enclosure	- Spraytight or in spraytight enclosures (hose test).
Rating	- To suit load requirements.
Operation	- Magnetic.
Type	- Across-line.
Duty	- Continuous.
Protection	- Low voltage release and overload protection.
Performance	- Automatic.

3.7.3.4 Reactivation (regenerator) electric heater. An electric reactivation heater shall be provided having nickel-copper alloy (monel) or corrosion-resisting steel sheaths and fins and tubular heating elements. The fins shall be bonded to the sheath to provide good heat transfer. The tubular heating elements shall be provided with hermetic end seals and shall be in accordance with MIL-H-22577. The heater shall be designed for continuous operation at the temperature required to meet the requirements of 3.3.2. The heating elements maximum surface temperature, with heater operating at rated voltage and rated air flow, shall not exceed 750°F., when the incoming air is at rated conditions.

3.7.3.4.1 Protection for heater. Each heater shall be provided with a protective thermostat to limit the maximum surface temperature of the heating elements to 900°F. or less, if there is reduction of air flow through heater. The thermostat shall be suitable for pilot duty at the heater voltage.

3.7.3.4.2 Heater connected load, maximum. At rated voltage the connected load for the electric heater shall not exceed 3.6 kw. for size 20-153, 6 kw. for the size 36-255 and 10 kw. for the size 55-505 dehumidifier.

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3.8 Welding. The surfaces of all parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. All welding shall be at least equal to that required by MIL-STD-278.

3.8.1 Where zinc-coated steel is used for fabricating parts, the metallic zinc shall be removed from all joints and surfaces on which welds are to be deposited and for a distance of 1/2 inch from the expected toes of the welds. In areas where the metallic zinc cannot be removed and it is necessary to weld over the zinc-coated surfaces, electrode type MIL-6010 in accordance with MIL-E-15599 shall be used.

3.9 Painting. All exterior surfaces of the dehumidifier shall be thoroughly cleaned and coated as follows:

(a) One coat of pretreatment coating in accordance with MIL-P-15328.
(b) One coat of zinc chromate primer in accordance with TT-P-645, or one coat of red lead primer in accordance with MIL-P-17545.
(c) Two coats of equipment enamel in accordance with class 2 of MIL-E-15090.

3.9.1 Interior surfaces. All interior zinc coated surfaces including welds and areas damaged by shearing, punching, bending, welding, or any other fabricated process, shall receive a thorough wash and a hot phosphate acide treatment, or a pretreating coating in accordance with MIL-P-15328, prior to applying at least two coats of primer conforming to TT-P-664 or MIL-P-17545.

3.10 Identification and label plates. Identification and label plates shall conform to types A, B, D, F or H of MIL-P-15024. Inscription or markings for all type plates shall be black except for warning plates which shall be red.

3.10.1 Size. In general, identification and label plates shall be of a size as listed in MIL-P-15024. Other sizes of plates may be used when specifically approved by the command or agency concerned.

3.10.2 Thickness. Thickness of the plates shall be in accordance with MIL-P-15024.

3.10.3 Application requirements. Each dehumidifier shall be provided with an identification plate and the following label plates.

(a) A warning plate: "Danger - 440 volts".
(b) A warning plate on each fan: "Check phasing before starting and rotation after starting."
(c) Direction of rotation on each fan and each motor.
(d) Humid air inlet.
(e) Conditioned air (dry air) outlet.
(f) Reactivation (regenerator) air inlet.
(g) Reactivation (regenerator) air outlet.
(h) Air filter cleaning requirements - at least every four weeks.
(i) Power leads location.
(j) Indication of the phase sequence A, B, C.

3.10.3.1 Other markings shall be provided where necessary, or where required by reference specifications.

3.11 Drawings. Drawings delineating the equipment shall be submitted in accordance with MIL-D-1000/2 except that type II drawings shall be approved by the Naval Ship Engineering Center or by the agency concerned. Type II drawings shall be furnished for each dehumidifier assembly and for the following component parts; fans, motors, heater and drive system. The drawings shall be in accordance with MIL-D-1000/2, and shall include information as required in this specification. Drawings shall indicate outline, mounting, attachment and connection dimensions including methods and sizes of fastenings and clearances for installation and servicing plus supplemental data as necessary to permit shipyard installation without the supplier's assistance. Drawings shall illustrate design, construction, identity of parts and total weight of assembled parts. Where necessary to show compliance with this specification, detail drawings shall be added to type II drawings.

3.11.1 Dehumidifier drawings. Type II drawings of the dehumidifier assembly shall cover the information specified in categories A, B, G, H and J of MIL-D-1000, including the following:

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(a) Clearances for withdrawal of parts or assemblies.
(b) Location, type and dimensions of humid air inlet, conditioned air (dry air) outlet, reactivation (regenerator) air inlet, reactivation air outlet, access panels, air filter access and electrical service entrance.
(c) Weight of each access panel.
(d) Arrangement showing component parts.
(e) Detail arrangement of desiccant contactor and air seals to show provisions for the prevention of by-passing air, and prevention of leakage of air between the reactivation and process portions of the desiccant structure.
(f) Center of gravity.
(g) All information necessary for preparation of foundation drawings including mounting plate details and drilling drawings with dimensions and tolerances.
(h) Schematic diagrams for electrical service, ductwork and air flow within the dehumidifier.
(i) Wiring diagram(s) of all electrical circuits (power and controls). All the electrical components shall be identified by a piece number of the bill of material in addition to a description or code identification and a detailed explanation of the operation of the controls shall be included. The wiring diagram shall clearly show the phasing and the power connections shall be identified for phase.
(j) List of label and instruction plates.
(k) Performance data, table form. The economy expressed in kilowatts per pound of water removal and the water rate removal expressed in pounds per hour shall be indicated for each entering air conditions, temperature in degrees Fahrenheit and percent relative humidity, specified in 4.3.3.2.
(l) Any special instructions for hoisting, alignment, initial lubrication, installation, or assembly, as necessary.

3.11.2 Fan drawings. Type II drawings of the fan shall cover the information specified in categories A, B, G, H and J or MIL-D-1000, including the following:

(a) Fan characteristic curves. The fan characteristic curves shall show total pressure at fan outlet from free delivery to no delivery, static pressure, brake horsepower required, and efficiency of the fan.
(b) Any special instructions for installation, alignment, removal, or initial lubrication, as necessary.

3.11.3 Motor drawings. Type II drawings of the motors shall cover the information specified in categories A, B, G, H and J of MIL-D-1000, including all the information required by the applicable motor specification specified herein.

3.11.4 Heater drawings. Type II drawings of the electric heater shall cover the information specified in Categories A, B, G, H and J of MIL-D-1000, including the following:

(a) Air volume - cubic feet per minute.
(b) Pressure drop.
(c) Kilowatt rating.
(d) Number of heat increments.
(e) Wiring diagram.
(f) Description and operation of protective thermostats and controls.
(g) Performance curves under varying entering air conditions.
(h) Any special instructions for removal, installation, or safety devices, as necessary.

3.11.5 Drive system drawings. Type II drawings of the drive system shall cover the information specified in Categories A, B, G, H and J of MIL-D-1000, including the following:

(a) Clearances and tolerances of moving parts.
(b) Arrangement of drive for the desiccant contactor (wheel).
(c) Detailed explanation of the operation of the system.
(d) Instructions for necessary adjustments.
(e) Any special instructions for alignment, initial lubrication, removal, or installation, as necessary.

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3.11.6 Wiring diagram. A reduced copy of the dehumidifier wiring diagram, either enclosed in a transparent water repellent envelope or printed on a plastic board, shall be permanently attached to the inside of a removable access panel.

3.11.7 Microfilm. Two sets of card-mounted microfilm for dehumidifier and component parts drawings shall be furnished in accordance with MIL-M-9868. The card-mounted microfilm shall be 35 millimeter (mm), filmed and processed in accordance with MIL-M-9868, with the following exceptions:

(a) Selection - Card mounted microfilm in accordance with type II, class 2 of MIL-M-9868.
(b) Size and weight of paper. In accordance with the aperture card size and document image location requirement specified in MIL-STD-804.
(c) Color - Buff.
(d) Information. Cards shall contain the information specified in MIL-STD-804 for engineering drawing card, Code A.
(e) The aperture cards (Card Code A) shall be coded in accordance with the requirements for engineering card (format A) of MIL-STD-804. Field (k), control activity, shall be punched "HR", fields (j) and (g) shall be left unpunched.
(f) Aperture opening. Location on card shall be in accordance with the aperture card and size and document image location requirements of MIL-STD-804.
(g) Submittal. After all drawings have been approved and notation of approval has been made, one complete set of card-mounted film shall be forwarded to each of the following:
(1) Naval Ship Engineering Center (6151B), Washington, D. C.
(2) Ships Parts Control Center, Mechanicsburg, Pa.

3.11.8 Roll microfilm. One roll of microfilm (or strip, if applicable) shall be furnished in accordance with MIL-M-9868/1.

3.11.8.1 Distribution of roll microfilm. After all drawings have been approved and notation of approval has been made, one set of roll microfilm shall be forwarded to the Naval Ship Systems Command for distribution to other Naval activies.

3.11.9 Certification data. Certification data drawing(s) for the dehumidifier shall be furnished for approval by the purchasing activity. The drawing(s) shall be in accordance with type III of MIL-D-1000/2.

3.12 Manuals. Manuals shall be furnished in accordance with type I of MIL-M-15071. Each size dehumidifier shall have its own individual manual. Reduced size copies of drawings and sketches plus photographs of the equipment to amplify and illustrate the text shall be included where applicable.

3.12.1 Notes, cautions and warnings. Notes, cautions, and warnings shall include, but shall not be limited to the following:

CAUTION: The wiring diagram shall be strictly adhered to in the assembly of the wiring.

3.13 Onboard repair parts. When specified (see 6.2), onboard repair parts shall be furnished in accordance with MIL-P-15137.

3.14 Workmanship. Workmanship shall be first class in every respect. Sharp edges, burrs and other imperfections shall be removed from all parts subject to contact with personnel to prevent cuts during repair and maintenance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Reliability and maintainability inspection. Reliability and maintainability inspection shall be in accordance with 4.2.1, 4.2.2 and 4.2.3.

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4.2.1 Production control and monitoring. The supplier shall provide and maintain an economical and effective system of product control and monitoring to assure that reliability achieved in design is maintained during production.

4.2.2 Reliability of subcontractor and vendor. The supplier shall provide an effective quality control system to assure that subcontractor and vendor selection and performance are consistent with the reliability requirements of the contract and applicable portions of this specification.

4.2.3 Maintainability demonstration. The first production unit shall be examined after testing, and the capability to maintain, disassemble and repair the unit shall be demonstrated to a Government representative. The demonstration shall be conducted utilizing the recommended tools and with other than expert mechanics. The maintainability demonstration shall include but shall not be limited to the following:

(a) Lubrication of bearings, bushings and gears without disassembly of the unit or any connections.
(b) Adjustment of drive mechanism (system) and air seals (when of a design which does not provide permanent sealing) without disassembly of the unit (see 3.3.6).
(c) Servicing of air filters without disassembly of the unit or any connections.
(d) Replacement of motor bearings.

4.3 Preproduction inspection. One sample of each size covered in the contract or order shall be submitted for preproduction inspection. Preproduction inspection shall consist of the examinations and tests specified in 4.3.3, 4.4.3 and 4.4.4. When high impact shock testing is specified in the contract or order, the samples shall be tested as specified in 4.3.2.

4.3.1 Unless otherwise specified (see 6.2), preproduction inspection shall not be conducted on dehumidifiers that are identical to those which have successfully passed the preproduction inspection within a period of 3 years of the date of the contract or order. Certified reports on such preproduction inspection shall be submitted in triplicate to the command or agency concerned to substantiate the fact that the tests have been conducted and specification requirements met.

4.3.2 High impact shock test (see 6.2). The dehumidifier shall be shock tested as prescribed for type A of MIL-S-901. The dehumidifier shall be operated at rated voltage and current during shock tests. The tests shall be conducted prior to tests specified in 4.3.3, 4.4.3 and 4.4.4. The correction of damage which may have occurred during shock tests shall not be performed prior to these tests. Evidence of fragmentation or missile effect of parts, deformation that will cause interference between parts or failure to operate shall be cause for rejection.

4.3.2.1 Upon completion of shock test and specified post-shock tests, the sample dehumidifier shall be returned to the supplier for his examination and correction of damage which may have occurred during shock tests prior to submitting sample dehumidifier for delivery.

4.3.3 Performance tests.

4.3.3.1 Capacity tests. The capacity of the dehumidifier shall be determined at the conditions specified in 3.3.2, 3.6.2 and 3.6.3.

4.3.3.1.1 Test procedures. The tests shall be conducted in a space in which the ambient and inlet temperature and humidity conditions can be maintained constant during the entire test. Capacity of dehumidifier shall be determined with the same condition of air entering the reactivation air inlet and conditioner air inlet. Provisions shall be provided for measuring and recording the inlet and outlet temperature and humidity, the power input, the rate of air flow, airflow total pressure at the outlets, temperature of air leaving electric heater and other necessary data. Prior to observation for record, the space shall be maintained at the test conditions and the dehumidifier shall be operated until equilibrium is established. Equilibrium shall be considered achieved when there is no significant difference in performance and the amount of water vapor removed from the steam of air to be dried shall be plus or minus 5 percent of the amount of water vapor gained by the reactivation air stream. Observations shall be made and recorded at 15 minute intervals for not less than one hour.

4.3.3.2 Economy and rate of water removal tests. The economy (kilowatts per pound of water removal) and the rate of water removal (pounds per hour) of each size dehumidifier shall be determined at each air condition (temperature and relative humidity) listed in table IV. The tests shall be conducted as specified

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in 4.3.3.1.1, and at least one test shall be conducted with the dehumidifier inclined 15 degrees from the normal horizontal position in any direction.

Table IV - Test conditions

Condition No.	Dry bulb temperature °F.	Relative humidity Percent
1	95	75
2	95	35
3	95	15
4	70	75
5	70	35
6	70	15
7	40	75
8	40	35
9	40	15
10	0	75
11	120	35

4.3.3.3 Sound test (see 6.2). The noise level of each size dehumidifier shall be established in accordance with the ASHRAE 36-62 under free delivery and air flow rates used for determining the capacity rating.

4.3.3.4 Motor tests. The insulation resistance test and the dielectric strength test shall be conducted as specified in MIL-M-17059 and MIL-M-17060, as applicable.

4.3.3.5 Weather test. The dehumidifier shall be subjected to the spray of a 3 inch fire hose for at least 15 minutes. The hose shall have a discharge pressure of at least 25 pounds per square inch gage and shall be directed at the equipment from a distance of not less than 10 feet nor more than 12 feet. All openings shall be blanked off for this test. Evidence of water inside of cabinet shall be cause of rejection.

4.4 Sampling for quality assurance inspection.

4.4.1 Lot. All dehumidifiers of the same size offered for delivery at one time shall be considered a lot.

4.4.2 Sampling for examination. Sample dehumidifiers shall be selected at random from each lot in accordance with table V for the examination specified in 4.4.3.

Table V - Sampling for examination

Lot size number of units	Sampler size number of units	Defective units	
		Acceptance number	Rejection number
1 to 6	All	-	-
7 to 15	7	0	1
16 to 25	10	0	1
26 to 40	13	0	1
41 to 65	17	1	2
66 to 110	22	1	2
111 to 180	28	2	3
181 to 300	35	2	3
301 to 500	45	3	4

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4.4.3 Examination. Each of the sample dehumidifiers selected in accordance with 4.4.2 shall be examined and measured to verify compliance with the requirements of this specification, not involving tests. Examination shall be conducted as specified in table VI. Any assembly in the sample containing one or more defects shall not be offered for delivery, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall not be offered for delivery.

Table VI - Classification of defects.

Categories	Defects
Critical:	
1	None defined.
Major:	
101	Size not as specified.
102	Incomplete, component parts missing (fans, heater, controls, safety devices, air filters).
103	Materials defective or not as specified.
104	Limiting dimensions, exceeded.
105	Welding incomplete, not free of cracks, nonfusion, heavy porosity, heavy undercut, slag inclusions.
106	Insulation, thermal, torn or missing.
107	Air seals, torn, missing.
108	Removable panel gaskets missing or defective.
110	Drawings not followed.
111	Blank flanges, inlets, outlets missing.
112	Bolts, nuts and screws not tight, missing (parts shall be properly fastened and secured).
113	Electrical connections not phased or loose.
114	Painting (as applicable), nonconforming.
115	Marking, manufacturer's name, not permanent, illegible or not as specified.
116	Label plates, not permanent, missing, illegible or not as specified.

4.4.4 Tests.

4.4.4.1 Operational test. Each dehumidifier furnished under a contract or order shall be energized to demonstrate proper construction. Safety devices and controls shall be checked to insure proper construction. Safety devices and controls shall be checked to insure proper performance, and insulation resistance and dielectric strength measurements shall be made to demonstrate that no personnel hazard exists. The dehumidifier shall be operated continuously for a 24 hour period at ambient temperature and humidity condition and corrected to test condition 5 of table IV to determine that the unit offered for delivery is equivalent to that which has been given approval under preproduction test. Evidence of improper construction, unsatisfactory performance of the safety devices and controls, personnel hazards or deficiency in capacity, shall be cause for rejection.

4.4.4.2 Circuit test. The equipment shall withstand for a period of 1 minute a dielectric test voltage of twice the rated voltage of the equipment plus 1000 volts. The frequency of the test voltage shall be not less than 60 cycles.

4.4.4.3 Air leakage test. Each dehumidifier furnished under a contract or order shall be subjected to an air leakage test. The air leakage test shall follow immediately the operating test, and before the cooling of the unit has taken place. A measured amount of air at not less than 3 inches of water static pressure difference between the process air and reactivation air circuits shall be introduced through a gas flow meter to the process air inlet with the process air outlet blocked off. The test shall be conducted with the desiccant contactor in motion and in a stationary position. The rate of leakage under the tests shall not exceed the value, determined under preproduction tests, required to meet the performance requirements of 3.3.2.1.

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4.4.4.4 Electric heater test. The insulation resistance test and dielectric strength test shall be conducted as specified in MIL-H-22577.

4.5 Reports. Certified reports on shock tests and preproduction tests shall be forwarded in triplicate to the command or agency concerned to substantiate the fact that the tests have been conducted and the specification requirements are met. Reports shall be assembled in manual form.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual orders.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Dehumidifier.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, size, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts.

5.1.2.1 Preservation and packaging. Onboard repair parts shall be cleaned, preserved and packaged by level A in accordance with MIL-E-16298. Items shall be packaged one item per unit package unless the item is used in sets.

5.1.2.2 Packing. Packing of onboard repair parts shall be as specified in 5.1.1.2.

5.1.2.3 Marking. The method of marking and the size of the marking of unit packages shall conform to MIL-STD-129. Shipping containers shall be marked as specified in 5.1.1.3.

5.1.3 Manuals. Manuals furnished with the basic equipment shall be packaged in sealed, waterproof containers.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging.

5.2.1.1.1 Level A. Each complete unit (see 3.3.1) shall be unit protected and packaged in accordance with method II of MIL-P-116.

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5.2.1.1.2 Level C. Preservation and packaging of each unit (see 3.3.1) shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity for early installation. This level may conform to the supplier's commercial practice.

5.2.1.2 Packing

5.2.1.2.1 Level A. Each complete unit, packaged as specified in the contract or order, shall be packed in a container conforming to any one of the following specifications at the option of the supplier:

Specification	Classification
PPP-B-601	Overseas type
PPP-B-621	Class 2
MIL-C-104	.....

Wood and wood-cleated boxes shall be provided with waterproof caseliners in accordance with MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Wood and wood-cleated boxes shall not exceed the weight limitations of the applicable box specification. Boxes shall be modified by the addition of wood skids in accordance with the applicable box specification. Blocking, bracing, anchoring and cushioning of container contents shall be in accordance with MIL-STD-1186 and the appendix to MIL-C-104. Shipping container closure shall be in accordance with the applicable box or crate specification or appendix thereto.

5.2.1.2.2 Level B. Each complete unit packaged as specified shall be packed in a container conforming to any one of the following specifications at the option of the supplier:

Specification	Classification
PPP-B-601	Domestic type
PPP-B-621	Class 1
MIL-C-104	.....

Wood or wood-cleated boxes shall be modified by the addition of skids in accordance with the applicable box specification. Wood or wood-cleated boxes shall not exceed the weight limitations of the applicable box specification. Blocking, bracing, anchoring and cushioning of container contents shall be in accordance with MIL-STD-1186 and the appendix to MIL-C-104. Shipping container closure shall be in accordance with the applicable box or crate specification or appendix thereto.

5.2.1.2.3 Level C. Each dehumidifier, packaged as specified in the contract or order, shall be packed in a container, at the lowest rates, in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the first receiving activity for immediate use. This level shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may be the supplier's commercial practice.

5.2.1.3 Repair parts. Repair parts shall be cleaned, preserved, and packaged as specified in 5.1.2.1 and shall be packed either level A, B or C as specified.

5.2.1.4 Manuals. Manuals shall be packed level C in accordance with MIL-M-15071.

5.2.1.5 Drawings. Drawings shall be packaged, packed and marked in accordance with MIL-D-1000/2.

5.2.1.6 Microfilm. Microfilm shall be packaged, packed and marked in accordance with MIL-M-9868/1.

5.2.1.7 Marking. In addition to any special marking specified, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The equipment covered by this specification is intended for Naval service where it is expected to withstand continuous use for long periods without benefit of overhaul. The equipment may

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serve vital spaces where failure would invariably result in serious reduction in the battle efficiency of the ship. In designing this equipment, the conditions under which the equipment will be maintained and repaired on shipboard and the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics should be considered. In designing this equipment for use on board a ship in commision, shock and noise requirements and use of service A motors should be considered.

6.1.1 The dehumidifier is intended for use in the preservation of inactive Naval ships, and for allied process on ships that are in commission.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Whether high impact shock testing is required (see 3.3.1.2 and 4.3.2).
(d) Whether noise treatment is required (see 3.3.1.3 and 4.3.3.3).
(e) Conditioned air fan requirements, if other than as specified in 3.6.2.
(f) Reactivation air fan requirements, if other than as specified in 3.6.3.
(g) Motor requirements, if other than as specified in 3.7.3.2.
(h) Whether controller is required (see 3.7.3.3).
(i) Onboard repair parts required (see 3.13).
(j) Preproduction inspection requirements, if other than as specified in 4.3.1.
(k) Selection of applicable preservation, packaging, packing and marking if other than as specified in 5.1 (see 5.2).

6.3 Design guides.

6.3.1 Shock. When high impact shock requirements are specified (see 3.3.1.2 and 6.2), the design information given in MIL-S-901 should be carefully considered in the design of this equipment. This information is also used by the drawing review activity as a factor in the approval of drawings prior to manufacture.

Preparing activity:
Navy - SH
(Project 4440-N0011)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS OVPRT 12-66)

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